

ODE Model of Gut Immune Response

1 State Variables

- $M_0(t)$: Naive macrophage population
- $M_1(t)$: Pro-inflammatory (M1) macrophage population
- $M_2(t)$: Anti-inflammatory (M2) macrophage population
- $P(t)$: Pathogen population
- $C(t)$: Commensal bacteria population
- $E(t)$: Epithelial health score ($0 \leq E \leq E_{\max}$)
- $R(t)$: Reactive oxygen species (ROS) concentration
- $D(t)$: Damage-associated molecular patterns (DAMPs)
- $A(t)$: Pathogen-associated molecular patterns (PAMPs)
- $S(t)$: Suppression-associated molecular patterns (SAMPs)

2 Auxiliary Variables

2.1 Signal Integration

$$\text{Danger} = D + A \tag{1}$$

$$\text{Safety} = S \tag{2}$$

$$I = 1 - \frac{E}{E_{\max}} \quad (\text{injury fraction, where } 0 = \text{healthy, } 1 = \text{fully damaged}) \tag{3}$$

2.2 Polarization Rates

$$r_{M_1} = \begin{cases} k_{\text{pol}} \cdot M_0 \cdot \frac{\text{Danger}}{\text{Danger} + K_{\text{pol}}} & \text{if } \text{Danger} \geq \theta_D \text{ and } \text{Danger} > \text{Safety} \\ 0 & \text{otherwise} \end{cases} \tag{4}$$

$$r_{M_2} = \begin{cases} k_{\text{pol}} \cdot M_2 \cdot \frac{\text{Safety}}{\text{Safety} + K_{\text{pol}}} & \text{if } \text{Safety} \geq \theta_S \text{ and } \text{Safety} > \text{Danger} \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

where k_{pol} is the polarization speed, K_{pol} is the half-saturation constant, θ_D is the danger threshold, and θ_S is the safety threshold.

2.3 Recruitment

$$r_{\text{recruit}} = \rho_D \cdot \text{Danger} \tag{6}$$

where ρ_D is the danger-dependent recruitment rate.

2.4 Microbe Leakage

$$L_P = \lambda_{P,I} \cdot I \cdot N_{\text{grid}} \quad (7)$$

$$L_C = \lambda_{C,0} \cdot N_{\text{grid}} + \lambda_{C,I} \cdot I \cdot N_{\text{grid}} \quad (8)$$

where $\lambda_{P,I}$ is the pathogen injury-leakage rate, $\lambda_{C,0}$ is the baseline commensal leakage rate, $\lambda_{C,I}$ is the injury-enhanced commensal leakage rate, and N_{grid} is the grid size.

2.5 ROS-Mediated Killing

$$K_{P,R} = \begin{cases} k_R \cdot (R - \theta_R^{(m)}) \cdot P & \text{if } R > \theta_R^{(m)} \\ 0 & \text{otherwise} \end{cases} \quad (9)$$

$$K_{C,R} = \begin{cases} k_R \cdot (R - \theta_R^{(m)}) \cdot C & \text{if } R > \theta_R^{(m)} \\ 0 & \text{otherwise} \end{cases} \quad (10)$$

where k_R is the ROS killing rate and $\theta_R^{(m)}$ is the ROS threshold for microbe killing.

2.6 Phagocytosis

$$\Phi_{\text{total}} = \alpha_{M_0} M_0 + \alpha_{M_1} M_1 + \alpha_{M_2} M_2 \quad (11)$$

$$f_P = \frac{P}{P + C} \quad (\text{pathogen fraction}) \quad (12)$$

$$f_C = \frac{C}{P + C} \quad (\text{commensal fraction}) \quad (13)$$

$$K_{P,\Phi} = \Phi_{\text{total}} \cdot f_P \cdot \frac{P}{K_{\Phi} + P + C} \quad (14)$$

$$K_{C,\Phi} = \Phi_{\text{total}} \cdot f_C \cdot \frac{C}{K_{\Phi} + P + C} \quad (15)$$

where α_{M_i} are the engulfment activities of each macrophage phenotype and K_{Φ} is the phagocytosis half-saturation constant.

2.7 Epithelial Dynamics

$$r_{\text{rec}} = \epsilon_{\text{rec}} \cdot (E_{\text{max}} - E) \quad (16)$$

$$D_{P \rightarrow E} = \kappa_P \cdot \frac{P \cdot E}{K_{P,D} + P} \quad (17)$$

$$D_{R \rightarrow E} = \begin{cases} \kappa_R \cdot (R - \theta_R^{(e)}) \cdot \frac{E}{E_{\text{max}}} & \text{if } R > \theta_R^{(e)} \\ 0 & \text{otherwise} \end{cases} \quad (18)$$

where ϵ_{rec} is the recovery rate, κ_P is the pathogen damage rate, $K_{P,D}$ is the pathogen damage half-saturation, κ_R is the ROS damage rate, and $\theta_R^{(e)}$ is the ROS threshold for epithelial injury.

2.8 Treg Activation

$$r_{\text{Treg}} = \begin{cases} \eta_{\text{Treg}} \cdot (M_1 + M_2) \cdot \frac{C}{K_{\text{Treg}} + C} & \text{if } \mathcal{K}_{\text{Tregs}} = 1 \\ 0 & \text{otherwise} \end{cases} \quad (19)$$

where η_{Treg} is the Treg activation efficiency, K_{Treg} is the activation half-saturation, and $\mathcal{K}_{\text{Tregs}}$ is a binary indicator for Treg presence.

3 Differential Equations

3.1 Macrophage Dynamics

$$\frac{dM_0}{dt} = r_{\text{recruit}} - r_{M_1} - r_{M_2} \quad (20)$$

$$\frac{dM_1}{dt} = r_{M_1} \quad (21)$$

$$\frac{dM_2}{dt} = r_{M_2} \quad (22)$$

3.2 Microbe Dynamics

$$\frac{dP}{dt} = L_P - K_{P,R} - K_{P,\Phi} \quad (23)$$

$$\frac{dC}{dt} = L_C - K_{C,R} - K_{C,\Phi} \quad (24)$$

3.3 Epithelial Health

$$\frac{dE}{dt} = r_{\text{rec}} - D_{P \rightarrow E} - D_{R \rightarrow E} \quad (25)$$

3.4 Signaling Molecules

3.4.1 ROS (Reactive Oxygen Species)

$$\frac{dR}{dt} = \beta_{R,M_1} \cdot M_1 \cdot \sigma_R - \delta_R \cdot R \quad (26)$$

where β_{R,M_1} is the M1 ROS production baseline activity, σ_R is the ROS addition rate, and δ_R is the ROS decay rate.

3.4.2 DAMPs (Damage-Associated Molecular Patterns)

$$\frac{dD}{dt} = I \cdot \sigma_D \cdot N_{\text{grid}} + \sigma_D \cdot \frac{P + C}{K_D + P + C} - \delta_D \cdot D \quad (27)$$

where σ_D is the DAMP production rate, K_D is the microbe-induced DAMP half-saturation, and δ_D is the DAMP decay rate.

3.4.3 PAMPs (Pathogen-Associated Molecular Patterns)

$$\frac{dA}{dt} = \sigma_A \cdot P - \delta_A \cdot A \quad (28)$$

where σ_A is the PAMP production rate and δ_A is the PAMP decay rate.

3.4.4 SAMPs (Suppression-Associated Molecular Patterns)

$$\frac{dS}{dt} = \sigma_S \cdot r_{\text{Treg}} - \delta_S \cdot S \quad (29)$$

where σ_S is the SAMP production rate and δ_S is the SAMP decay rate.

4 Parameter Summary

4.1 Initial Conditions

$$M_0(0) = N_{\text{phago}} \quad (30)$$

$$M_1(0) = 0 \quad (31)$$

$$M_2(0) = 0 \quad (32)$$

$$P(0) = 0 \quad (33)$$

$$C(0) = N_{C,0} \quad (34)$$

$$E(0) = E_{\max} \cdot \left(1 - \frac{\text{injury}_{\%}}{100} \cdot \frac{L_{\max}}{L_{\max}}\right) \quad (35)$$

$$R(0) = D(0) = A(0) = S(0) = 0 \quad (36)$$

where $E_{\max} = N_{\text{grid}}^2 \cdot 6$ (maximum health score), N_{phago} is the initial phagocyte count, $N_{C,0}$ is the initial commensal count, $\text{injury}_{\%}$ is the initial injury percentage, and $L_{\max}$ is the maximum injury level.

4.2 Parameters from ABM (Sampled via LHS)

- θ_D : Danger activation threshold (activation_threshold_danger)
- θ_S : Safety activation threshold (activation_threshold_SAMPs)
- β_{R,M_1} : M1 ROS production (activity_ROS_M1_baseline)
- $\alpha_{M_0}, \alpha_{M_1}, \alpha_{M_2}$: Engulfment activities (activity_engulf_*_baseline)
- ϵ_{rec} : Epithelial recovery chance (epith_recovery_chance)
- $\theta_R^{(m)}$: ROS threshold for microbe killing (th_ROS_microbe)
- $\theta_R^{(e)}$: ROS threshold for epithelial injury (th_ROS_epith_injury)
- $\lambda_{P,I}, \lambda_{C,0}, \lambda_{C,I}$: Leakage rates (rate_leak_*)
- ρ_D : Danger recruitment rate (recruitment_rate_danger)
- $\sigma_R, \sigma_D, \sigma_A, \sigma_S$: Signal production rates (add_*)
- $\delta_R, \delta_D, \delta_A, \delta_S$: Signal decay rates (*_decay)

4.3 ODE-Specific Parameters (Require Calibration)

- $k_{\text{pol}} = 0.1$: Polarization speed
- $K_{\text{pol}} = 0.5$: Polarization half-saturation
- $\kappa_P = 1.0$: Pathogen damage rate
- $\kappa_R = 2.0$: ROS damage rate
- $K_{P,D} = 50$: Pathogen damage half-saturation
- $k_R = 2.0$: ROS killing rate
- $K_{\Phi} = 100$: Phagocytosis half-saturation
- $\eta_{\text{Treg}} = 0.01$: Treg activation efficiency
- $K_{\text{Treg}} = 50$: Treg activation half-saturation
- $K_D = 100$: DAMP microbe half-saturation

4.4 Fixed Parameters

- $N_{\text{grid}} = 25$: Grid size
- $N_{\text{phago}} = 125$: Initial phagocyte count (0.2×25^2)
- $N_{C,0} = 20$: Initial commensal count
- $E_{\text{max}} = 3750$: Maximum epithelial score ($25^2 \times 6$)
- $\text{injury}_{\%} = 60$: Initial injury percentage
- $L_{\text{max}} = 5$: Maximum injury level

5 Notes

- All rates are per timestep (dimensionless time units matching ABM)
- Michaelis-Menten kinetics used throughout for saturation effects
- System is deterministic (no stochastic terms)
- Well-mixed assumption (no spatial structure)
- Initial conditions match ABM initialization where possible